

- **Electric Drives project**

PID Voltage Control for Switched Reluctance Motor using Zeta Converter

- **Team Members:**

#	Name	ID
1	Ahmed Mahmoud Mohamed Aly Mahfouz	40-19285
2	Mohamed Mahmoud Sayed Shelkamy Ali	40-18924

- **Evaluator:**

Examiner: Dr. Walid Atef Omran

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Contents

Abstract	3
Keywords.....	3
1 Introduction.....	3
2 Literature Review.....	5
3 Simulink Model	6
3.1 Switched Reluctance Motor Model.....	7
3.2 Zeta Converter Model.....	7
3.3 Voltage Control Model	8
4 Results and Discussion	10
5 Conclusion and Future Recommendations	23
6 References	24

- **Abstract**

The aim of this project is to control the speed of a Four-phase Switched Reluctance Motor using a PID controller to control the output voltage of an AC to DC ZETA converter. The controller was created as two cascaded PID controllers one controls the output voltage of the zeta converter and the other PID controller is to control the filtered current of the AC supply. A Simulink model was developed and the simulation results were obtained to test the functionality of the developed system.

- **Keywords**

SRM; Zeta Converter; PID controller; Speed Control;

- **Introduction**

The switched reluctance motor (SRM) has some advantages compared to the induction motor (IM) and permanent magnet synchronous motor (PMSM), such as a magnet-free and robust rotor structure that allows high speed operations and is inherently fault tolerant. Due to these reasons, SRM is a promising traction machine candidate for electric vehicle(EV) applications. The SRM's principal drawbacks include excessive torque ripple, low torque/power density, and low power factor. When considering the needs of traction applications, torque ripple minimization is one of the most significant and hardest SRM design factors. The SRM drive is a variable-speed motor. High-efficiency, high-speed drive. Its rotor is clear of any debris. Compared to other available magnets, windings, and brushes, traditional motors It's easy to use because of its straightforward design. cost-effective for mass production.

In SRM, the stator and the rotor have floating poles. The set of windings wound on the stator terminals 6/4, 8/6, 12/10 and 12/6 is among the commonly available configurations. At any time, opposite pairs of poles are energized, and the rotor poles try to line up in the direction of the path of least resistance. At times, the other two pairs of rotor poles remain unchanged for the time being. The continuous excitation of motor phases leads to an actual torque of in the forward direction and therefore to a motor rotation of. The cyclic change in resistance at the periphery of the motor causes torque to develop. DC excitation should be introduced in the increased inductance region during regeneration, and the phase currents should be synchronized with the reduced inductance configuration. To allow the motor to operate in the constant torque region in the range low speed by limiting Current of phases using fixed frequency PWM control or hysteresis control is used. The current of the phases is controlled so that the diminishing residual current does not cause any negative torque in motor mode.

The torque in the SRM does not depend on the direction of the current in the phases. Therefore, for SRM control, single-pole converters are well suited compared with AC machines that require bidirectional current. Mutual coherence between phases is negligible in SRM, which is beneficial because each phase winding is independent in terms of control and torque generation. Although one of the disadvantages of the lack of mutual coupling is that the stored magnetic energy must be handled with care if necessary, thus a path is provided for the field energy during phase transition single. The power converter topologies available for SRM control are unbalanced converter, C-dump converter, 2-wire converter, and mid-range converter. The midrange converter is a low cost converter compared to other converter topologies available because it only requires one active solid state switch per phase. This mid-range converter consists of a two-series series capacitor as a DC link. Therefore, the voltage of the capacitor must be balanced for the proper functioning of this inverter. Since the speed can be easily adjusted by simply varying the intermediate circuit voltage.

The torque generated is a nonlinear function of the phase current and the position of the rotor. The non-linearity of the torque is mainly due to the construction of the floating pole and to the magnetic saturation. Excitation of the single phase coil of the SRM at a time is one of the common methods of operation, resulting in increased torque ripples and hence speed ripples which are reflected as shown. Radial force is developed in the rotor due to torque and speed ripples resulting in vibration and acoustic noise. Many control techniques are cited in the literature to reduce torque ripple.

Improving energy quality in today's world is one of the indispensable requirements. The IEC 61000-3-2 power quality standard recommends high power factor (PF) and low total harmonic distortion (THD) on the power supply side. The traditional SRM drive scheme powered by a diode bridge rectifier (DBR) draws non-sinusoidal current from the AC grid, resulting in 70% to 80% of high current THD on the AC grid. Therefore, it is strongly recommended to use a power quality converter to reduce total harmonic distortion (THD) and obtain a high power factor (PF). Due to the quality improvement of the AC grid and the control of the DC link voltage, these single-stage converters have attracted people's attention.

Literature Review

Self-reluctance motors become more and more attractive for many industrial applications due to their simple structure, high speed ,high torque but it has a main disadvantage which is the torque ripple. Here we will show a number of papers that control the switched reluctance motor using different techniques.

In paper [1] A method to improve SRM digital current control is proposed. This proposed PI current control method includes gain adaptation as a function of incremental inductance and compensation for EMF. Modifying the PI current control allows to minimize the disturbances in the control loop , which are related to saturation or resistance to EMF and variable time constants. This document focuses on SRM with low current, because the incremental inductance in this case has a great change and has a significant negative impact on the dynamics of the controller. experimental results show that the current controller can perform high dynamic output changes, and the closed-loop gain can be maintained at a constant high value, so as to achieve good dynamic behavior in the entire operating range. This improved current control can be used effectively to control torque.

This paper[2] describes the selection and development of voltage-based control strategies to achieve average torque control of multiple SRMs in HEV applications. The SRM and controller perform as expected, are acoustically quiet, and provide the kind of response and stability the driver expects. The aggressive timing of the two cars The schedule and extensive advertising promises dictated the use of simple control strategies. The memory and computation requirements of this voltage-based control strategy are very low, so you can use a basic 8-bit microcontroller and a simple programmable logic device to run the , or use the cheap DSP,If higher torque accuracy is required and the dynamic requirements of the are moderate, a solution could be the , using the voltage controller described in this article as the feedforward combined with the rms current control loop (or current). Similarly.

This paper[3] presents a proposal for a new type of regulator for switched reluctance motor (SRM) drives. The proposed regulator enables a significant extension of the rotational speed range and drive output power. This regulator is characterized by a complex structure, including two regulation modules: voltage and phase supply switch-on angle. The voltage module includes a proportional integral derivative (PID) voltage regulator. During its operation, the value of the phase supply switch-on angle and the width of the phase supply range are determined as a result of interpolation of the data previously determined in the simulation program. The other module contains the PID

controller of the phase supply switch-on angle. The values of the angles included in the tables have been determined so as to ensure that the drive works with the greatest possible efficiency. The control method is determined based on the current operating parameters of the drive, i.e., torque and speed. The operation of the regulator was simulated in the MATLAB Simulink program. The regulator presented here was implemented in a field programmable gate array (FPGA). Tests of the regulator's operations in the prototype system were carried out in the field of control of commutation angles.

- **Simulink Model**

Figure 1 shows the Simulink model of the project including the SRM model, the Zeta Converter, and the PID Control loop.

The shown configuration includes a reluctance-reluctance motor (SRM) driver with a front PFC converter. The recommended SRM drive uses a midpoint converter to power the SRM. IGBT switches and diodes are used to power each phase. The SRM drive speed is adjusted by changing the DC link voltage. Use voltage sensor Control the speed of the proposed drive and the current sensor used for reactive power compensation in the AC network. When using this driver, good motor drive characteristics can be achieved. The proposed SRM driver is designed to operate within a wide range of power supply voltage and a wide speed control range, while improving the quality of the AC network with low power supply current THD and high power factor. Due to its simple design, SRM is a viable alternative to AC machines, thereby reducing machine costs under the same category. [4]

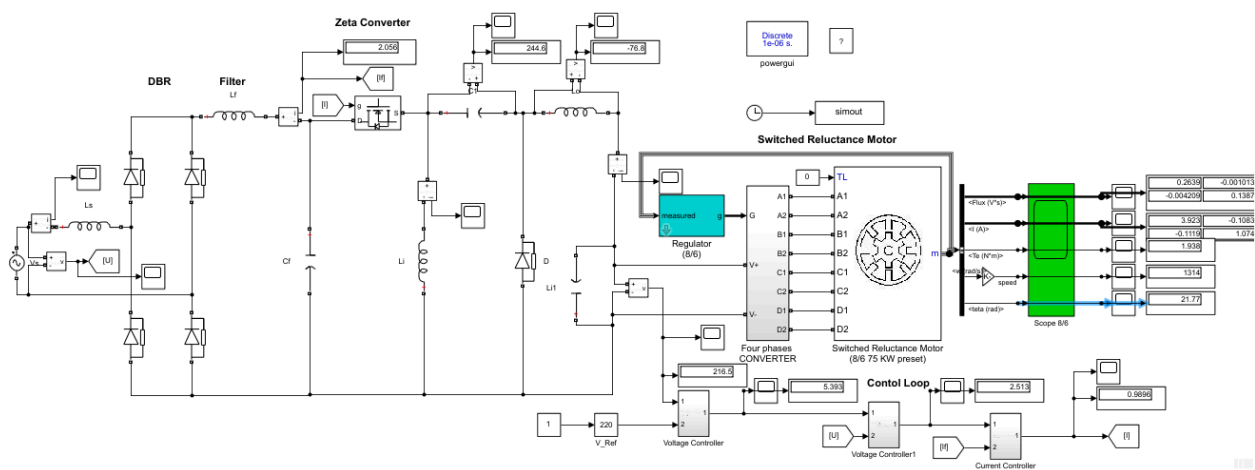


Figure 1: The project Simulink Model

1. Switched Reluctance Motor Model Description

Figure 2 shows the Model of the Switched Reluctance Motor (8/6 75 KW).

The SRM is powered by a four-phase asymmetric converter with four branches, and each branch is composed of two IGBTs and two freewheeling diodes. When turned on, the active IGBT applies a positive power supply voltage to the stator winding to drive the positive current in the stator winding. Phase winding. During winding, a negative voltage is applied to the winding, and the accumulated energy is returned to the constant current source through the diode. In this way, you can reduce the edge steepness of the current in the motor windings. Through the sensor connected to the rotor, the angle of the motor phase can be accurately set. Through these switching angles, the generated torque waveform can be controlled. The phase currents are independently controlled by three hysteresis regulators generated by IGBTs. These regulators adjust the signal by comparing the measured current with the reference current. The switching frequency of the IGBT is mainly determined by the hysteresis band [5].

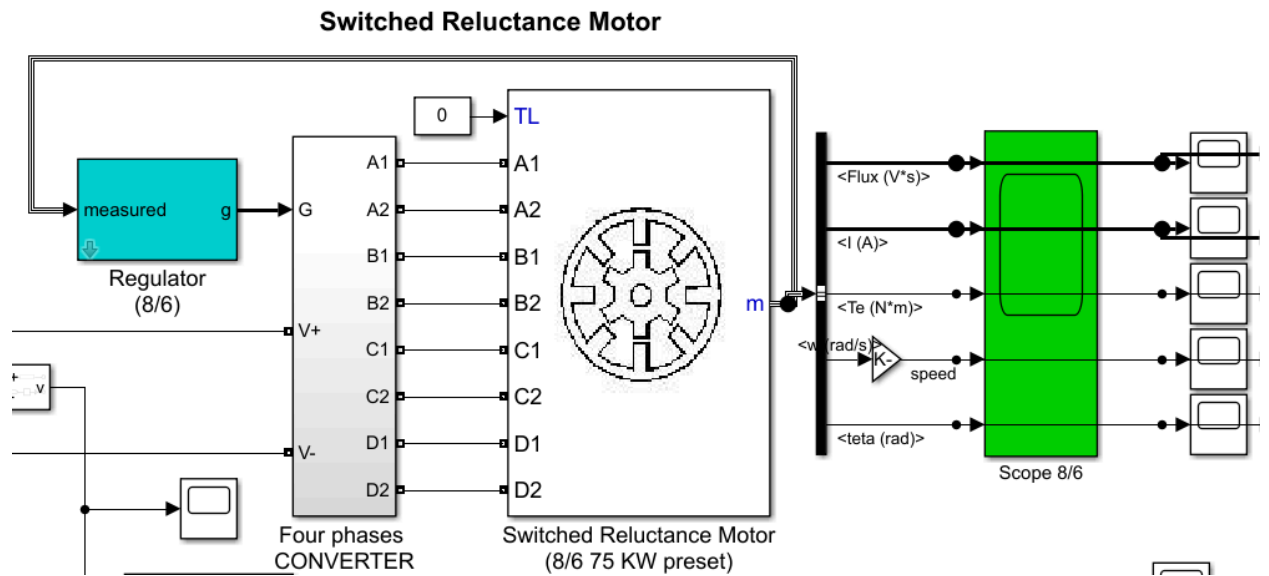


Figure 2: The SRM Simulink Model

2. Zeta Converter Model Description

Figure 3 shows the Simulink model for the Zeta Converter used to drive the Motor.

The shown Zeta-PFC is designed to operate in CCM (continuous conduction mode) so that the input inductor current (i_{Li}), output inductor (i_{Lo}) and the voltage on the intermediate capacitor C1 are used in continuous conduction mode (CCM) Switch completely under switching cycle. The Zeta converter works in two different modes. The two different operating modes are as follows:

Mode I: When the switch Sw is turned on, the source provides current to the inductor Li , and the capacitor $C1$ discharges through the output inductor Lo , so the input inductor current i_{Li} and the output inductor current i_{Lo} increase linearly.

Mode II: During this operating mode, the switch Sw is turned off and the diode D conducts current, so Li and Lo can transfer their energy to the intermediate capacitor $C1$ and the output capacitor Co , respectively. In this mode, there is no source current in the circuit, which reduces the THD in the line current [2].

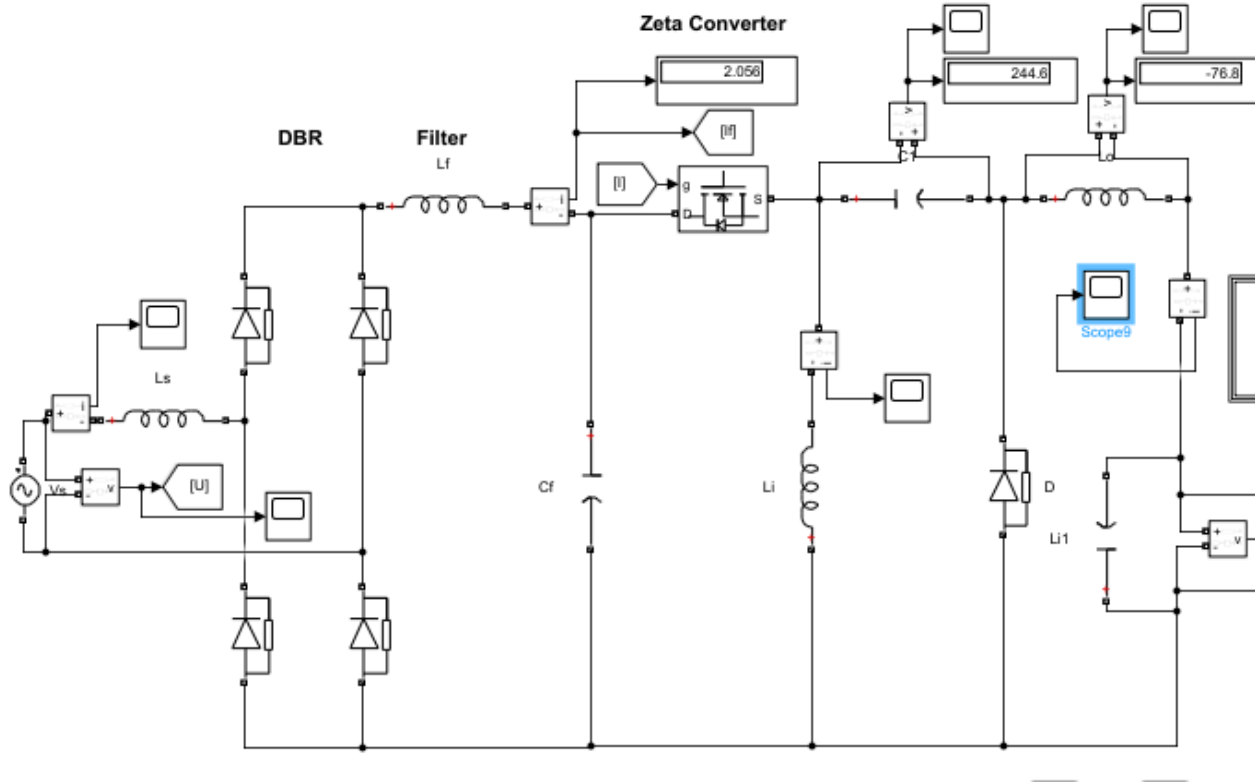


Figure 3: The ZETA Converter Simulink Model

3. PID Control loop

Figure 4 shows the Simulink model for the PID control loop used to Control the Motor

1. Control of Front-End PFC Converter

The PFC operation of the SRM driver includes the control of the input current of the converter and the DC output voltage of the converter. The SRM drive uses the front PFC converter, and the diode rectifier bridge is connected after the AC power supply. Its output is fed to Zeta. PFC operation and speed change by changing the inverter output voltage are achieved by changing the inverter's turn-on time (D). Inverter fixed switch (fs) The inverter voltage is distributed between two

capacitors connected in series with the center point N. The overall control includes the development of control algorithms, selection of sensors to determine current and voltage, and circuit breaker switching pulses to obtain the required PFC speed control.

2. Control of SRM

For SRM, the switching sequence of the midpoint converter is provided by a position sensor that detects the rotor position θ . The turn-on and turn-off angles are functions of the phase inductance, which depend on the position of the rotor. The ignition angle and switching angle can be continuously adjusted by a fixed retard angle. This depends on the number of rotor poles and phases. Here, we consider an 8/6 pole SRM with four stator phases. According to the inductance distribution, the ideal signal positions of the four-phase SRM are A phase $0-15^\circ$, B phase $15^\circ-30^\circ$, C phase $30^\circ-45^\circ$ and $45^\circ-60^\circ$. For phase D, the four-phase switching sequence is calculated based on this assumption[4].

The control loop consists of two cascaded PID controller as the following:

-Voltage controller: It controls the error between the output DC voltage of the converter and the reference voltage and generates a current signal to be fed into the current controls.

-Current controller: It controls the error signal between the current signal generated by the voltage controller and the filter current and creates an output feedback signal to be fed to the gate of the power electronics component of the Zeta converter which is the MOSFET to control the converter DC output voltage.

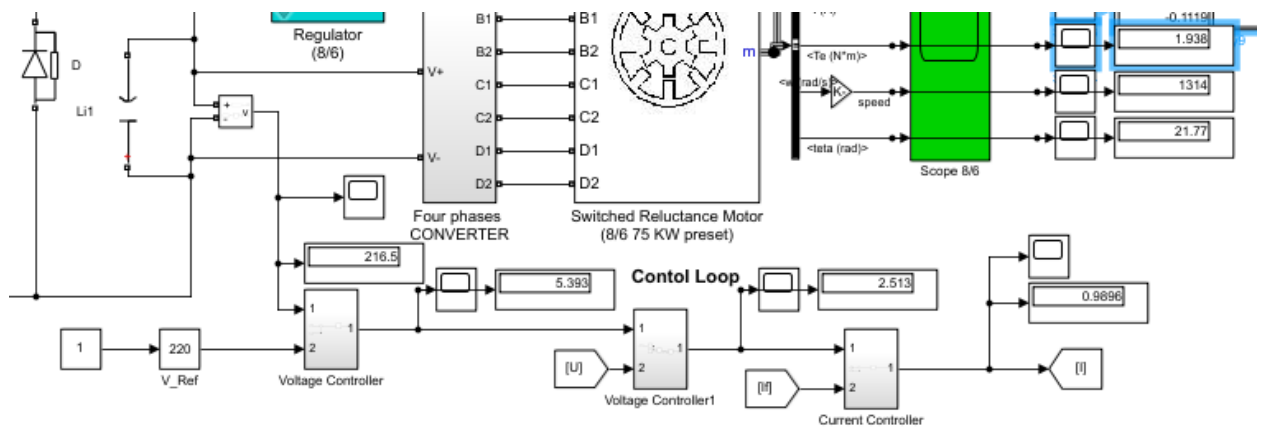


Figure 4: The PID Control Simulink Model

- Results and Discussions

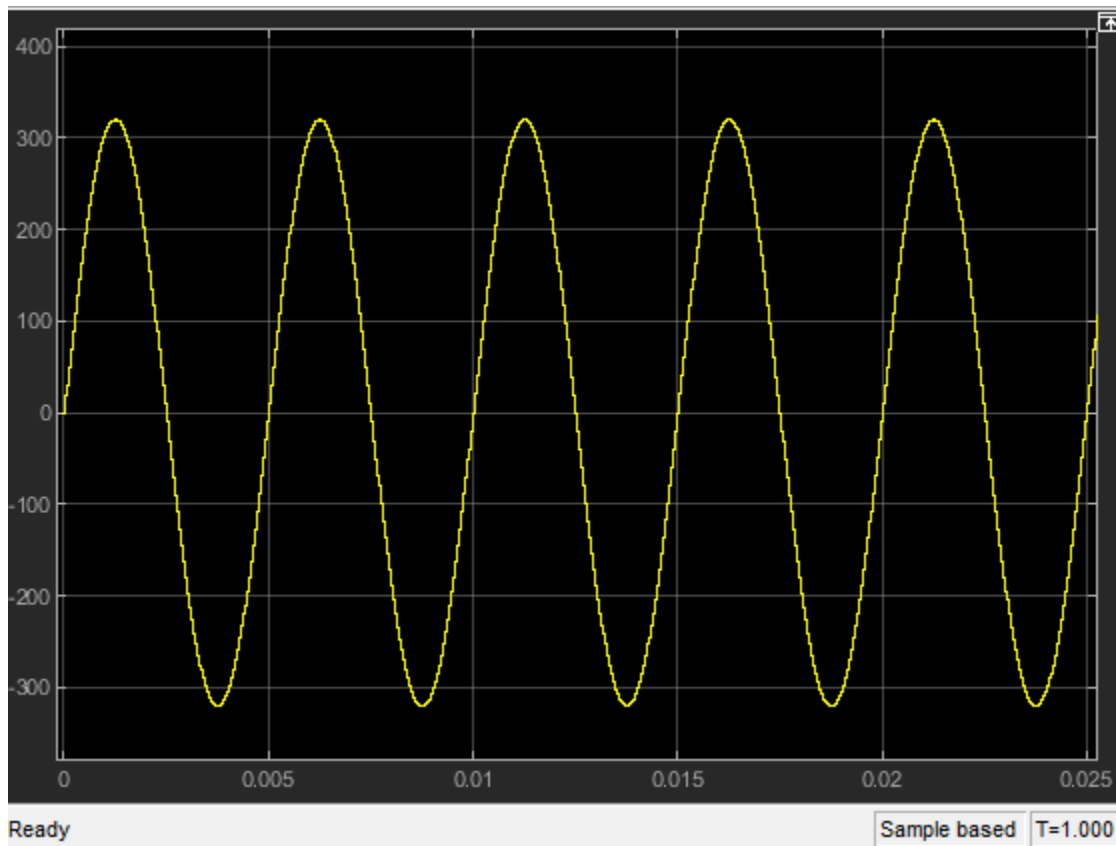


Figure 5: AC Power supply $V_s=320\text{V}$ and $f_s=200\text{Hz}$

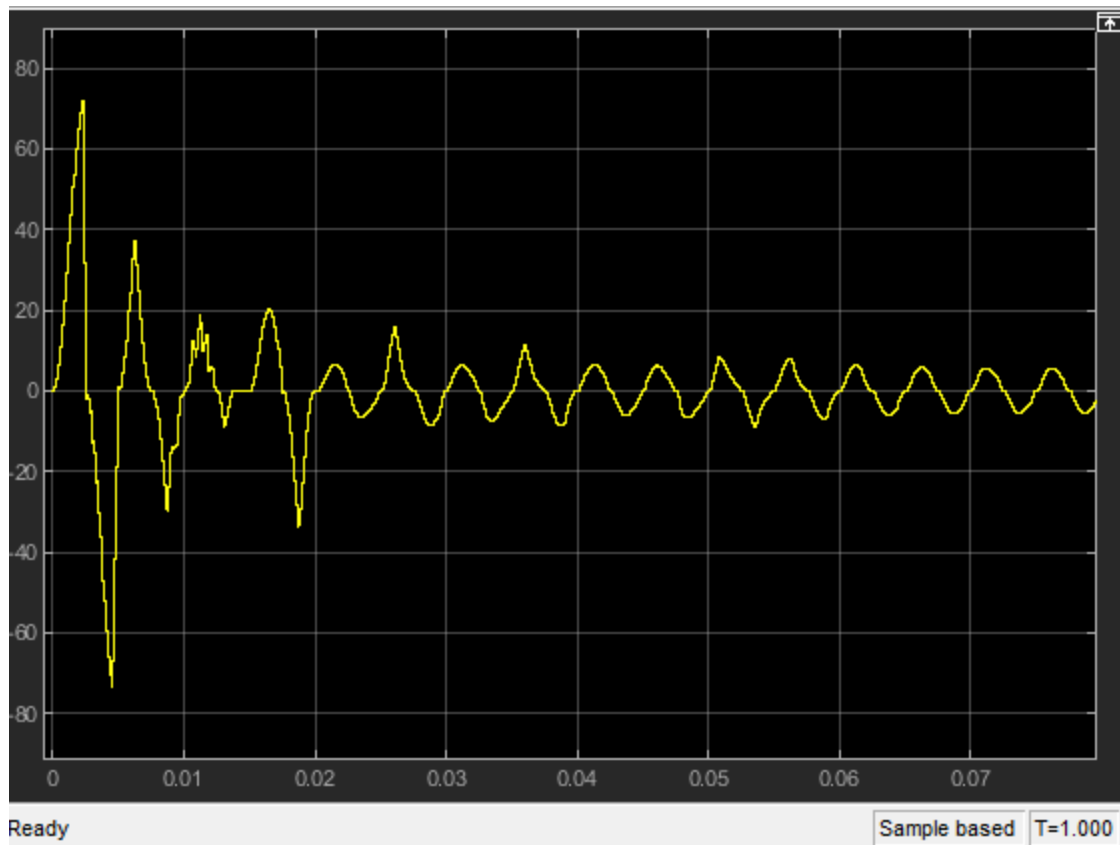


Figure 6: Power Supply current i_s

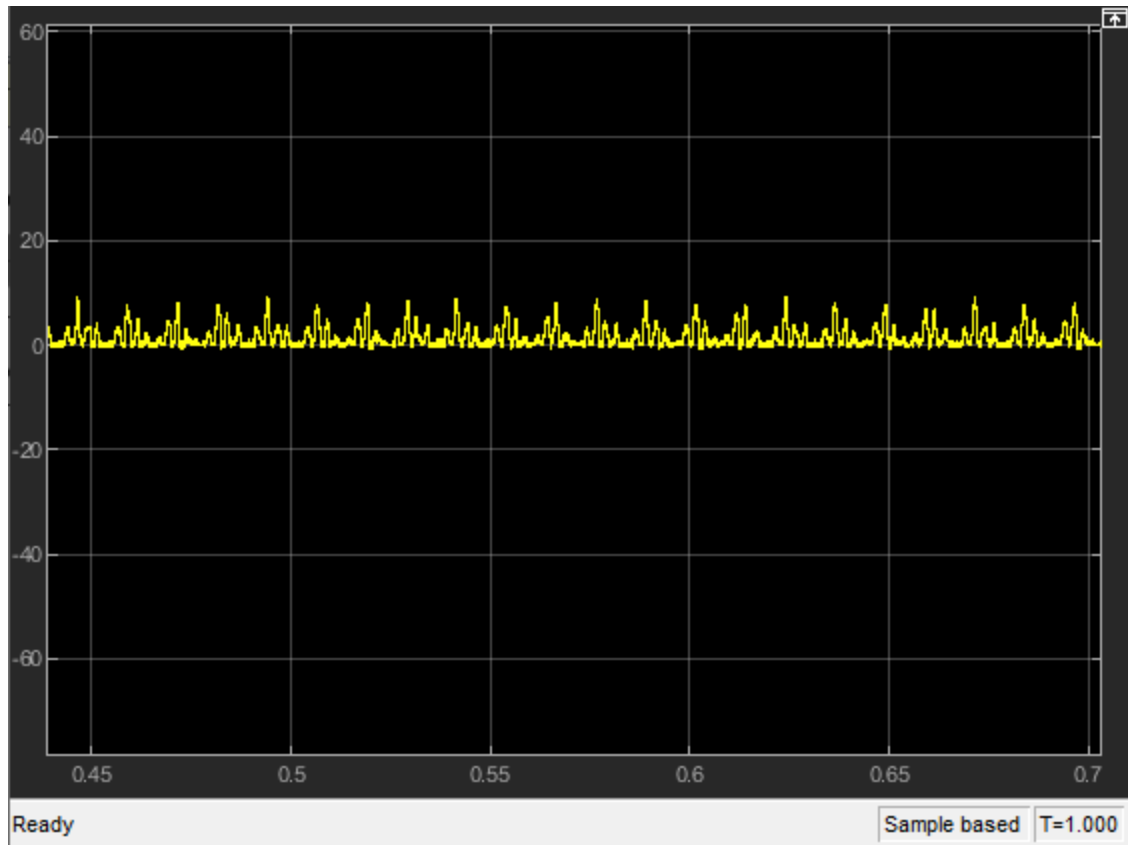


Figure 7: Power Supply current i_{Li}

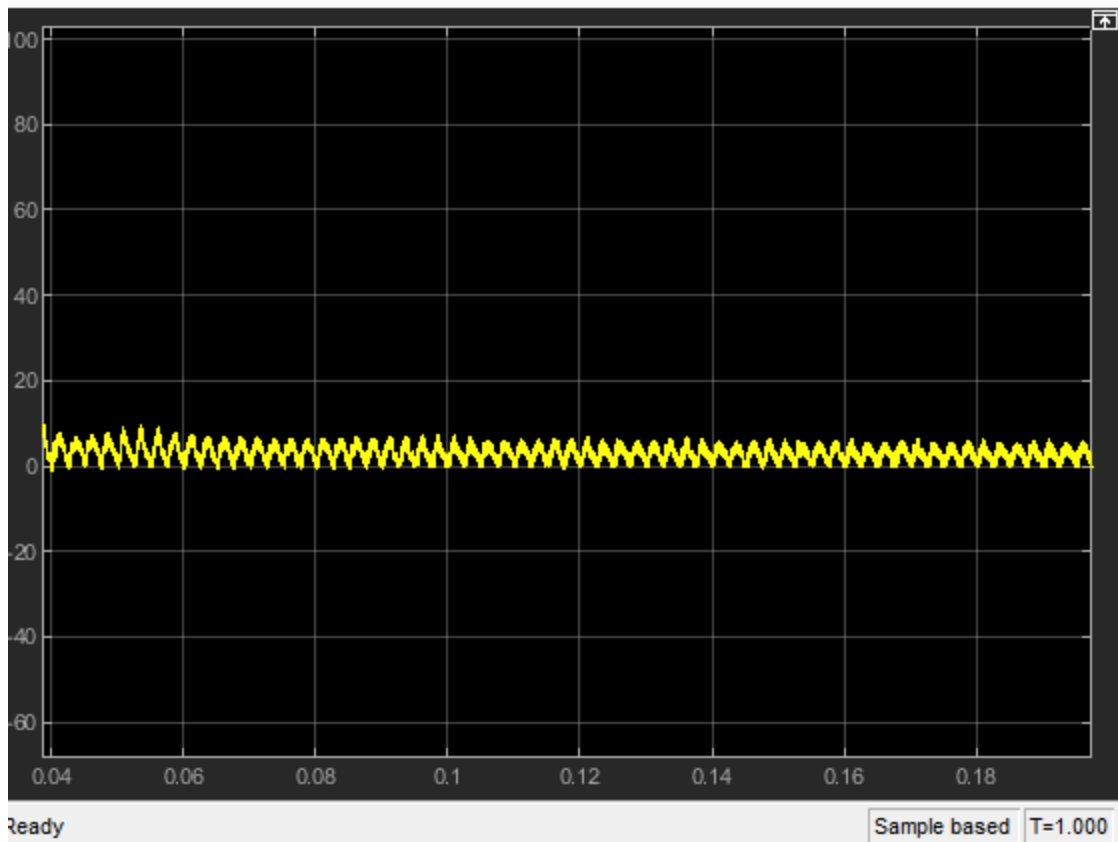


Figure 8: Current of the input inductor i_{L_o}

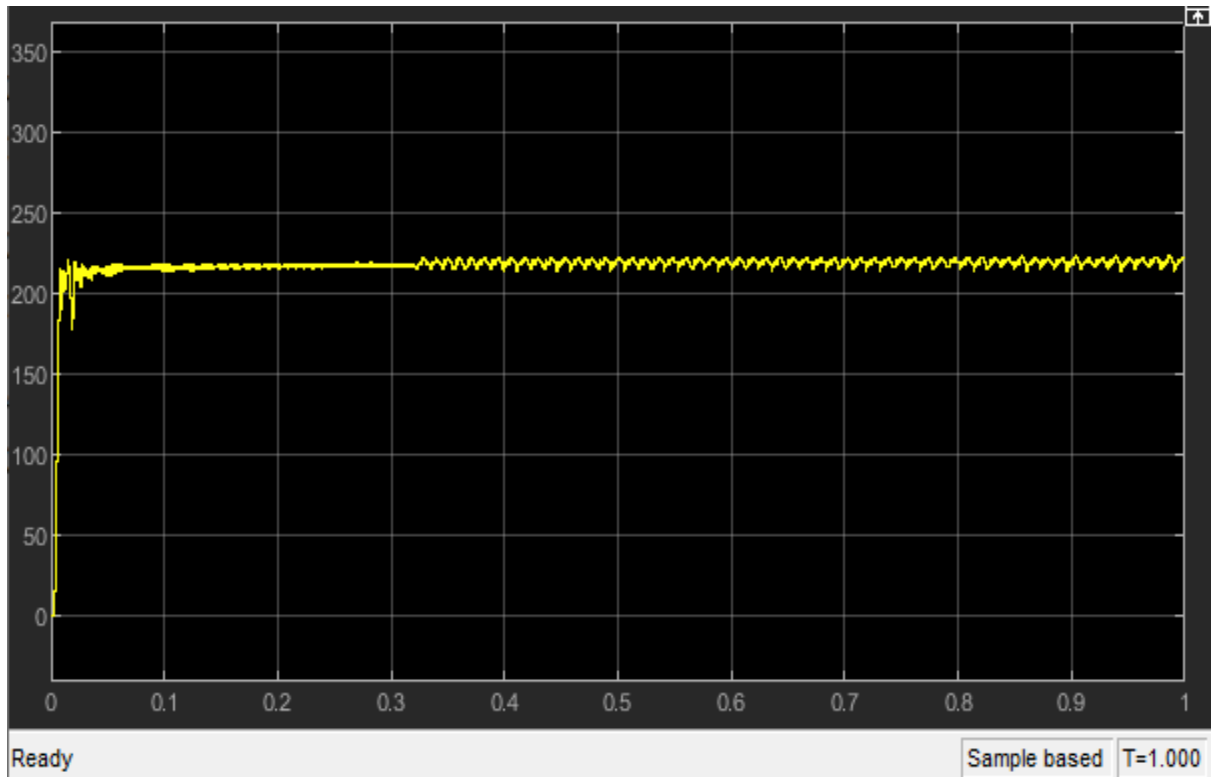


Figure 9: Converter DC output voltage at $V_{\text{ref}}=220$ V

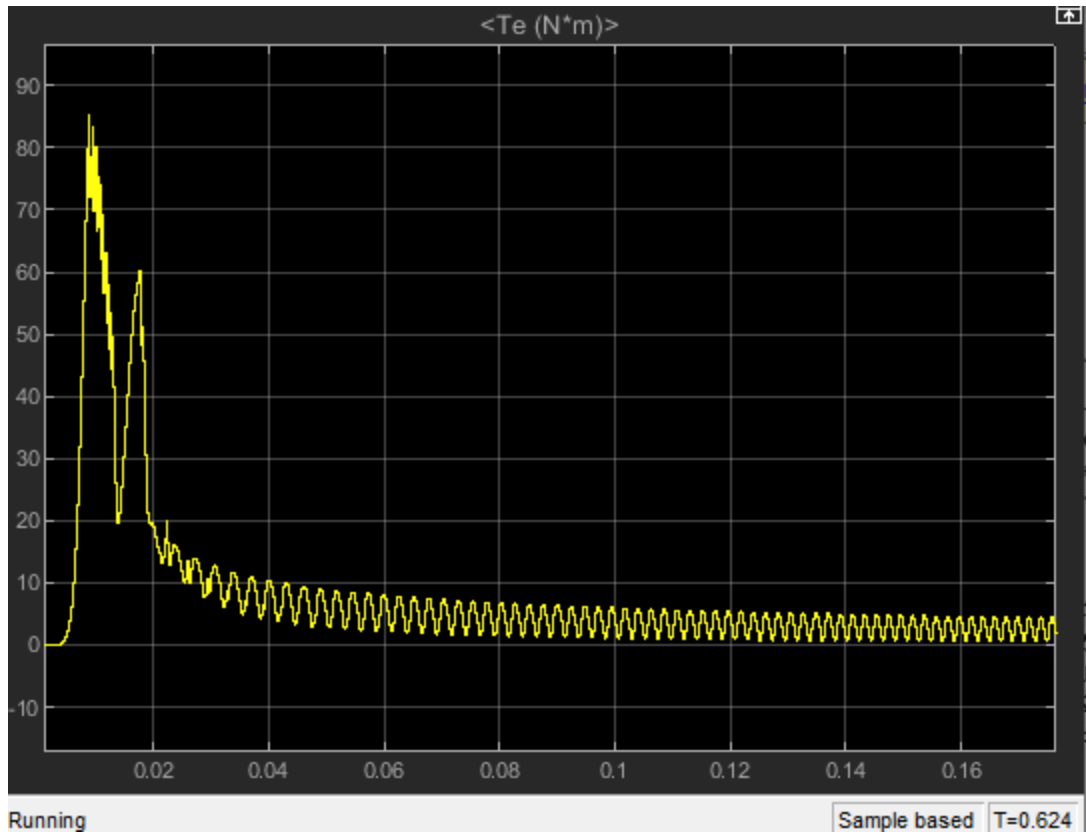


Figure 10: Motor output torque T_m

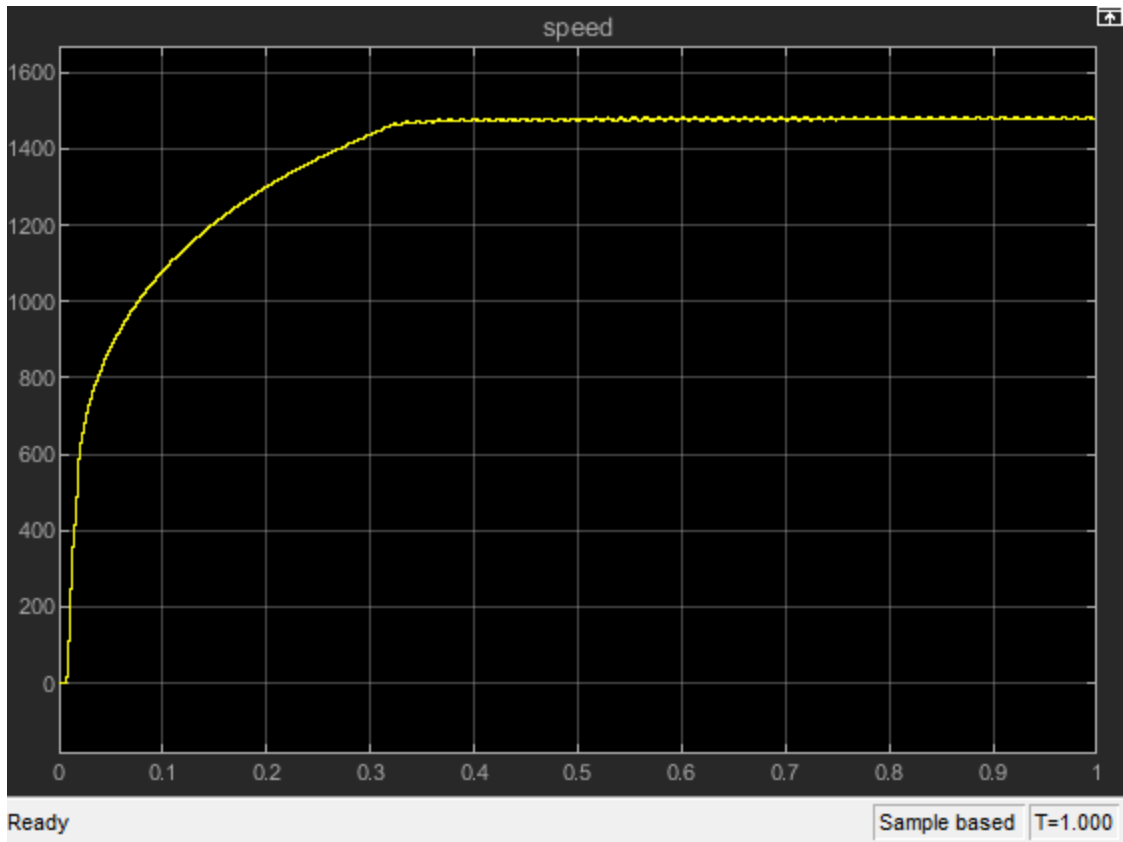


Figure 11: Motor Output speed N_M

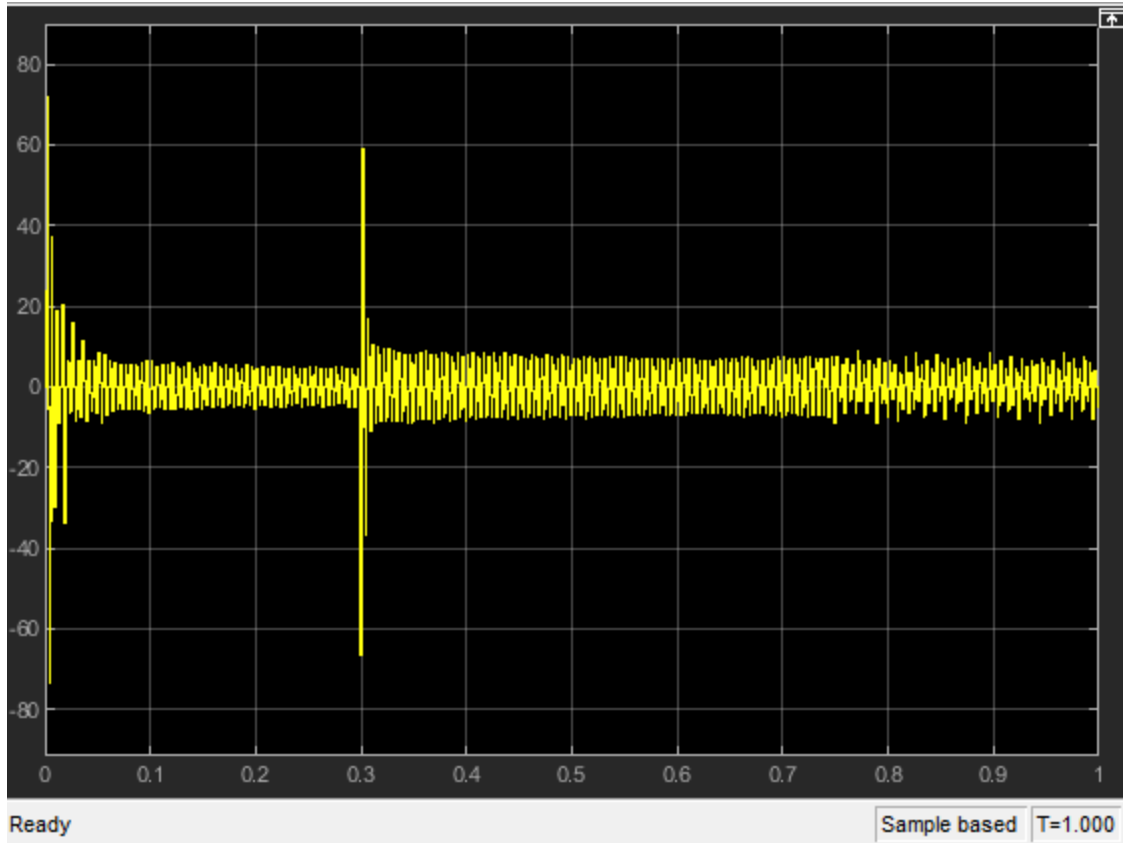


Figure 12: Power Supply current i_s when V_{ref} changed from 220V to 320V at ($t=0.3$ s)

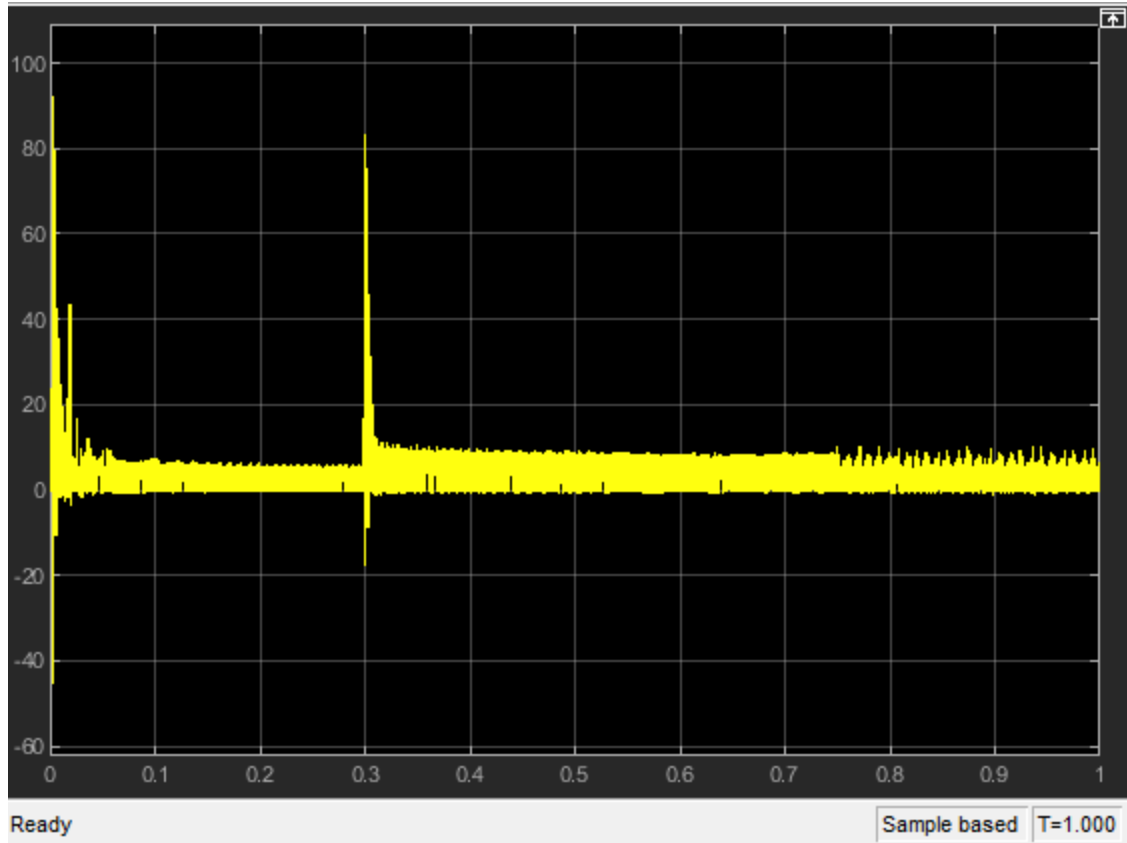


Figure 14: Input inductor current i_{Li} when V_{ref} changed from 220V to 320V at ($t=0.3$ s)

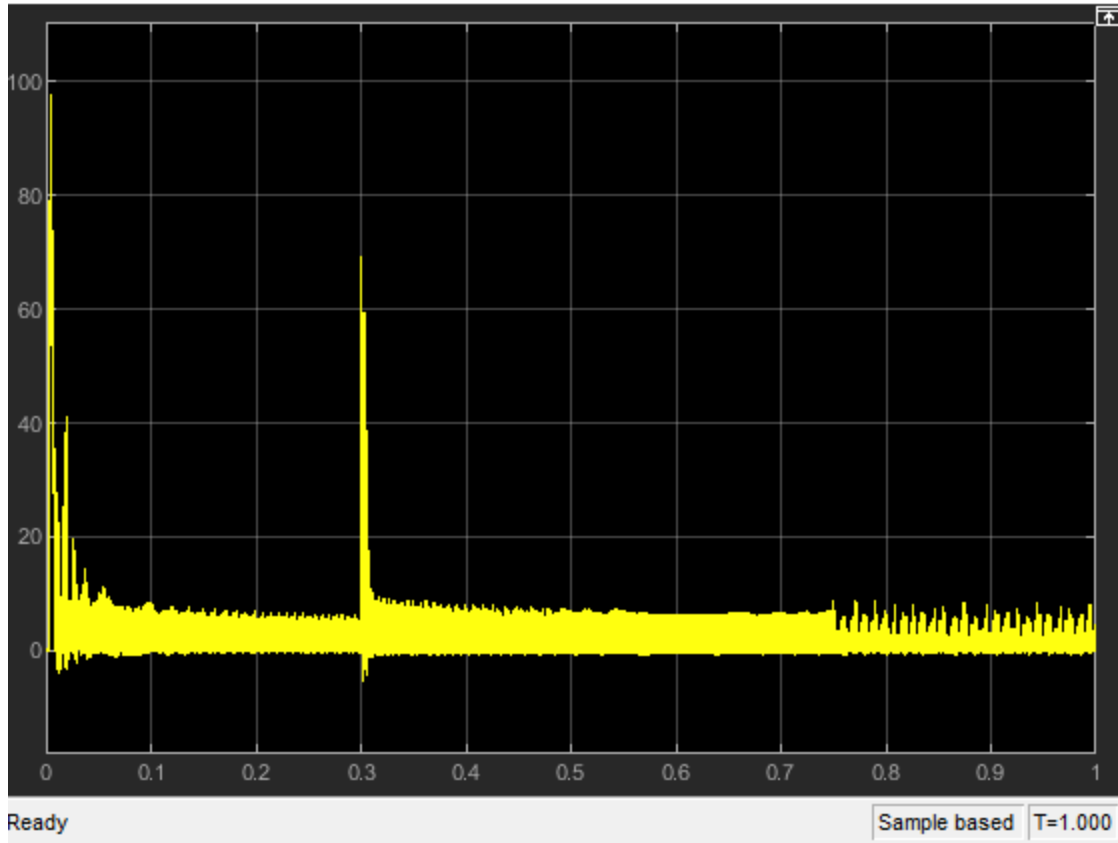


Figure 15: Output current i_{L_o} when V_{ref} changed from 220V to 320V at ($t=0.3$ s)

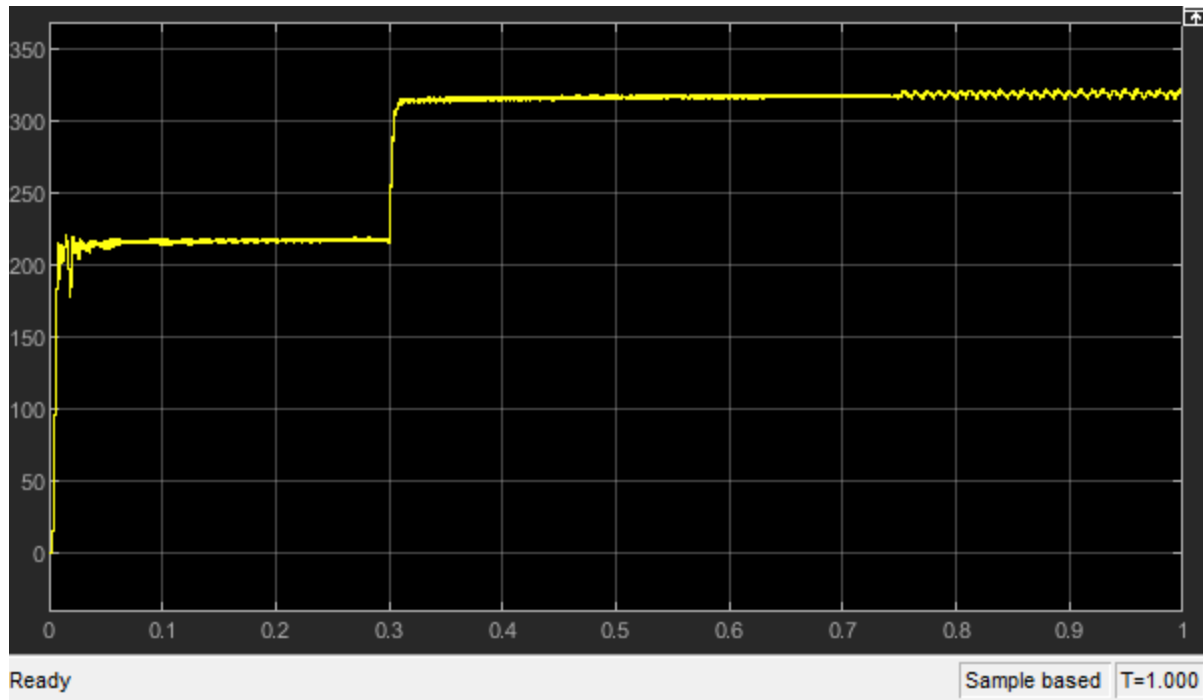


Figure 16: Converter output voltage V_{dc} when V_{ref} changed from 220V to 320V at ($t=0.3$ s)

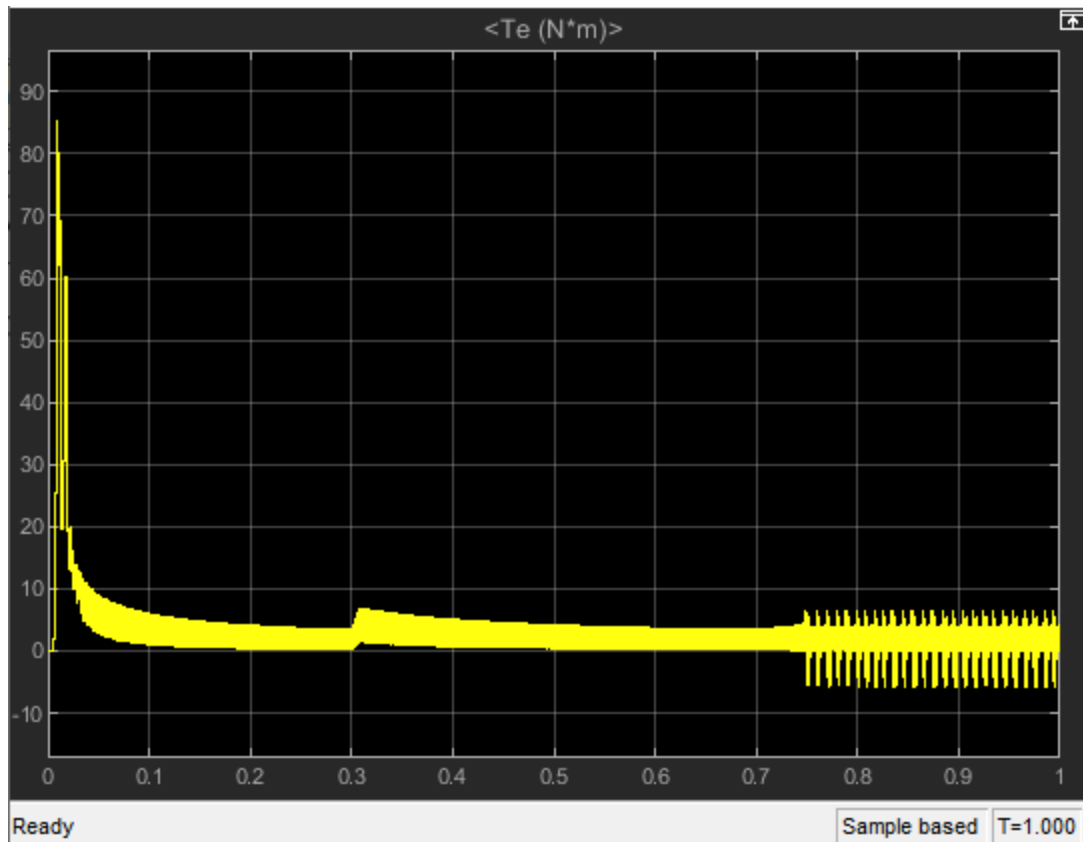


Figure 17: Motor output torque T_m when V_{ref} changed from 220V to 320V at ($t=0.3$ s)

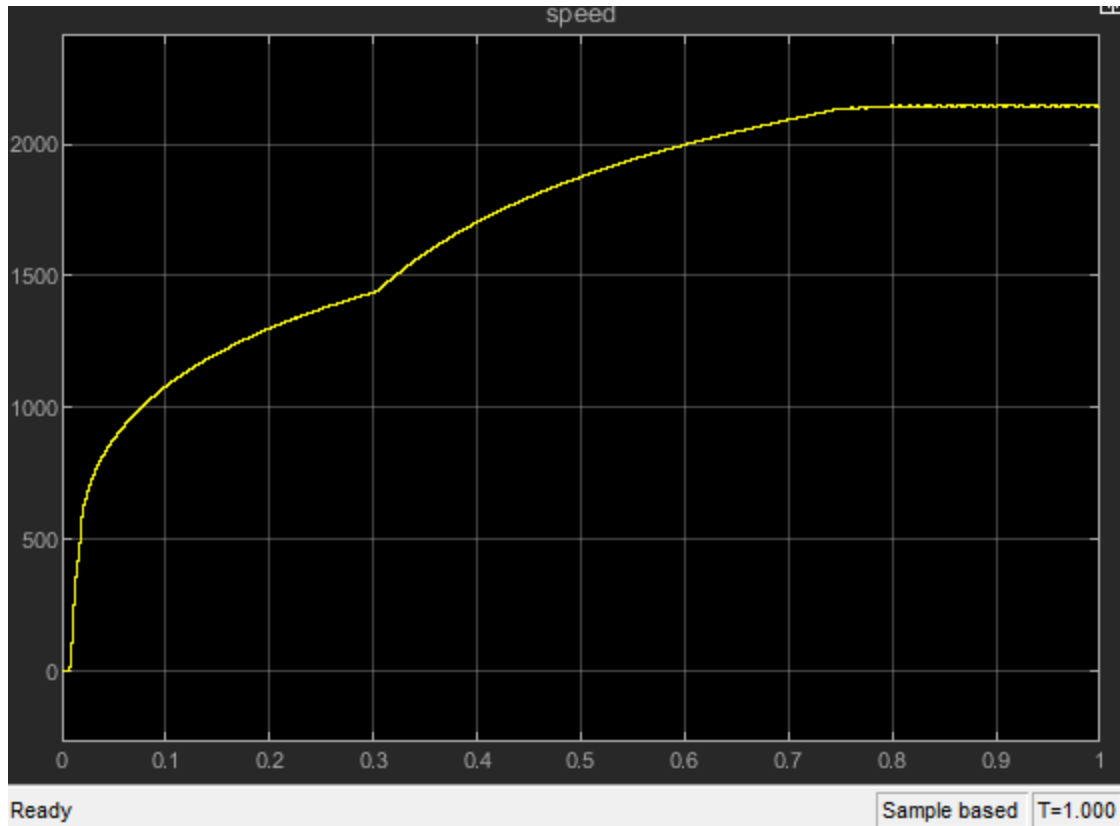


Figure18:Motor output speed T_m when V_{ref} changed from 220V to 320V at ($t=0.3$ s)

The results obtained are shown in Figures 6-18 are used in analysis to improve AC mains power quality. Figure 6-11 shows the steady-state operation of the proposed SRM drive for a DC link voltage of 220 V with a supply voltage of 320 V. The resulting supply current is essentially sinusoidal and is phase with the supply voltage reflecting the unity power factor from the AC mains. Since the zeta converter is designed with CCM, it is possible to obtain continuous current of intermediate inductor L_i and output inductor L_o . The motor operates with a variable DC link voltage and the rated speed of 1450 rpm is obtained with a DC link voltage of 320 V as shown in Figure 11. Figure 12-18 shows the dynamic behavior during a step change at a DC link voltage. The DC link voltage increased from 220V to 320V, and the performance of the drive during this step change of the DC link voltage is well shown in the figure. The performance of the controller is satisfactory as the smooth adjustment of the DC link voltage is obtained.

- **Conclusion and Future Recommendations**

PFC-based Zeta converter-supplied SRM motor drive, was proposed for a wide range of speeds. The PFC Zeta converter was designed to work with CCM and act as a unique power factor correction. The drive of the proposed SRM motor was designed to be simulated in a MATLAB / Simulink environment to achieve power factor correction with an AC power supply and to control speed over a wide speed range. Satisfactory performance drives have been achieved, and as a result, the has been proven as the recommended solution for low voltage applications.

- **References**

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